

Research Findings and Analysis Report - DataGear

Heading

DataGear: Data Protection Education

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Research Memo Draft

Introduction

In the digital age, we rely on computers and online systems to store, access, and manage our personal and professional data. However, this method comes with significant risks because data can get lost, compromised, or stolen due to inadequate security measures, human error, or cyber threats. Without proper management and awareness, individuals may unknowingly expose their information to malicious third parties, possibly leading to identity theft, financial loss, or unauthorized account access. Moreover, many users—especially those without a strong technical background—are unsure how their information is protected or unaware that they may be oversharing data online. Even users with good intentions often feel overwhelmed by privacy decisions or confused by contradictory information.

Problem Statement

It is imperative to educate individuals who are less familiar with IT security on how they can enhance their data protection, recognize potential threats, and adopt safe online behaviors.

Research Question

We aim to understand how users perceive and manage their digital privacy and whether cybersecurity-related education can improve their ability to recognize and respond to digital threats.

Audience

Our project, DataGear, is focused on educating the general public—particularly non-experts—on best practices for digital privacy and cybersecurity. We identified and engaged with several key audiences during our initial research phase. Our primary audience includes everyday technology users, especially individuals who fall into younger (Gen Z) or older (Boomer) age groups. These users often feel overwhelmed, uninterested, or unsure about how to protect themselves online. Based on our research, we developed personas such as the overwhelmed beginner—someone who struggles with navigating privacy settings and understanding basic digital protections, regardless of age—and the curious but misinformed

user—someone who is interested in learning more but often follows unsafe advice or has incomplete knowledge. So far, we have engaged with 22 Gen Z (75.9%), 1 Millennial (3.4%), 5 Gen X (17.2%), and 1 Boomer II (3.4%) users through our survey (Figure 1). Additionally, we surveyed a small number of individuals who rated themselves as technological experts. We engaged 2 (6.9% of respondents to the technical proficiency question) experts during our research, meanwhile, all other respondents rated themselves as having an advanced or intermediate technological proficiency. Furthermore, we still hope to connect with additional stakeholders such as digital educators and cybersecurity professionals to gain valuable perspectives not just from within the industry but from those actively shaping the educational cybersecurity landscape. We also seek to survey older adults with low digital literacy who could feel potentially intimidated or avoid technology altogether.

Methodology

Online surveys were conducted to collect data. The team worked together to develop quantitative questions and a qualitative prompt. A total of twenty-four quantitative and one qualitative questions were developed. One example of a quantitative question we asked is: “Do you use any technologies that employ zero-trust techniques?” For our qualitative prompt, we included: “Briefly describe the biggest challenge you face in protecting your personal data online.” All of our questions and prompts have the intention of clarifying the malware experience of the participants to hopefully better craft a solution. Each team member gathered at least five participants representing a range of generations (Gen Z, Millennials, Gen X, Boomers II, Boomers I).

Analysis

Our project incorporates both quantitative and qualitative approaches to our data analysis. We imported our survey results into Power BI to extract data-driven insights from three visualizations we created. First, we created a decomposition tree (Figure 2) to explore how low cyber-awareness (<2) scoring respondents vary across generations and cybersecurity-related behaviors. We calculated this awareness score on a numeric 0-5 scale, with scores of 5 being the most cyber-aware respondents. Essentially, the score is calculated as such:

- **+1** if the respondent is aware of data protection rights.
- **+1** if they avoid using public Wi-Fi for sensitive tasks.
- **+1** if they read privacy policies (always or sometimes).
- **+1** if they know about zero-trust technologies.
- **+1** if they avoid misinformation (i.e., did **not** believe incognito = full anonymity) and selected at least one correct belief.

This calculated score, while limited in its comprehensiveness since we cannot ask follow-up questions from surveys, allows us researchers to measure cybersecurity awareness for the purposes of our project. Additionally, while we cannot attach a Power BI file to this

document, Figure 2 displays an example of one fully drilled-down awareness breakdown path of the data. For example, looking down the Gen Z respondent route, it seems that many people who use public wifi for sensitive data transactions do not have an antivirus, VPN, or read website privacy policies but sometimes create backups. Although not all of these data elements are always used to make an insight at the same time, this visualization helps us look at associations between different cybersecurity behaviors. We can see that our respondents are not making significant efforts to protect themselves despite sharing sensitive data over public wifi networks. Despite some protective measures being taken in some cases, respondents still seem to take minimal precautions online. To further illustrate these patterns across a broader set of respondents and behaviors, Figure 3 presents a bar chart summarizing key cybersecurity actions through our calculated awareness score, allowing us to compare how often specific practices occur across different demographic groups.

Figure 3 is a bar chart broken up in small multiples highlighting self-reported proficiency levels versus calculated awareness scores among age groups. Looking at the chart, many Gen Z respondents rate themselves as advanced technically, but this group also overlaps with low awareness scores. There is an apparent gap between confidence and awareness where the respondents' perceived proficiency does not equate to secure digital behavior. Gen Z was the only generation that spanned across all calculated awareness levels, indicating a high level of variance in their perceived skills. Additionally, respondents appeared reluctant to identify as experts in digital security practices, highlighting a disconnect between the availability of cybersecurity knowledge and users' willingness to engage with it. Further, Figure 4 is a matrix looking into data protection rights awareness by generation. The majority of respondents (27/29) are either unsure or do not know their data protection rights, further supporting the lack of foundational cybersecurity knowledge across generations. While these quantitative visualizations offer a broader picture of trends across respondent groups, our qualitative analysis provides context to the behavior patterns through open coding, axial coding, and affinity diagram methods.

To better understand these behaviors, we also conducted a qualitative analysis of the open-ended survey responses using open coding (Figure 5). We identified recurring keywords and expressions such as “confusing,” “overwhelming,” “don’t know what’s protected,” and “it’s too much effort,” which signaled deeper emotional and cognitive barriers to safe digital practices. These codes were then grouped through axial coding (Figure 6) into three overarching themes: Information Overload, Gaps in Knowledge & Accessibility, and Awareness & Critical Evaluation. These themes backed up our quantitative findings and helped our persona development and design priorities. To build on these insights, we developed an affinity diagram (Figure 7) that visually grouped and ordered common user behavior patterns. For example, we observed that overwhelmed users often skip verification steps and use public Wi-Fi without protection, while curious but misinformed users may incorrectly believe that browsing in incognito mode keeps them anonymous and often lack antivirus software.

Recognizing common themes in users keywords and behaviors can help guide our research by allowing us to focus more on the psychological and behavioral parts of decision making and how emotional barriers like anxiety can impact user engagement. For website design, our three themes we mentioned earlier will serve as a roadmap for enhancing and

simplifying user experience. For example, information overload seems to be a conflict of interest, so we would prioritize creating concise and short content that gets to the point. Using simpler language for beginners and more visual aids with explanations or clickable elements would address the knowledge gap. Our last theme of critical evaluation can be fostered by giving users real life scenarios to address risks and concerns so that they can make informed decisions. Overall, these insights we've found can not only help guide our design, but also tailor it to user needs.

Data Visualizations

[Link to Power BI visualization PDF](#) to see visualizations in better quality

Figure 1

What generation are you categorized in?
29 responses

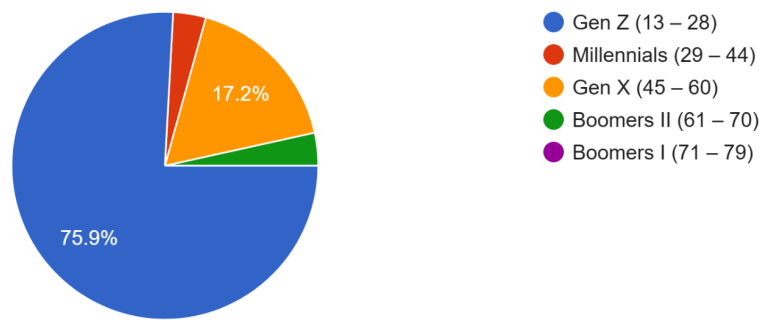


Figure 2

Breaking Down Low Cyber-Awareness Scores

By Generation, Public Wifi Usage, Anti Virus Usage, VPN usage, Reading a Website Privacy Policy, and Frequency of Backups

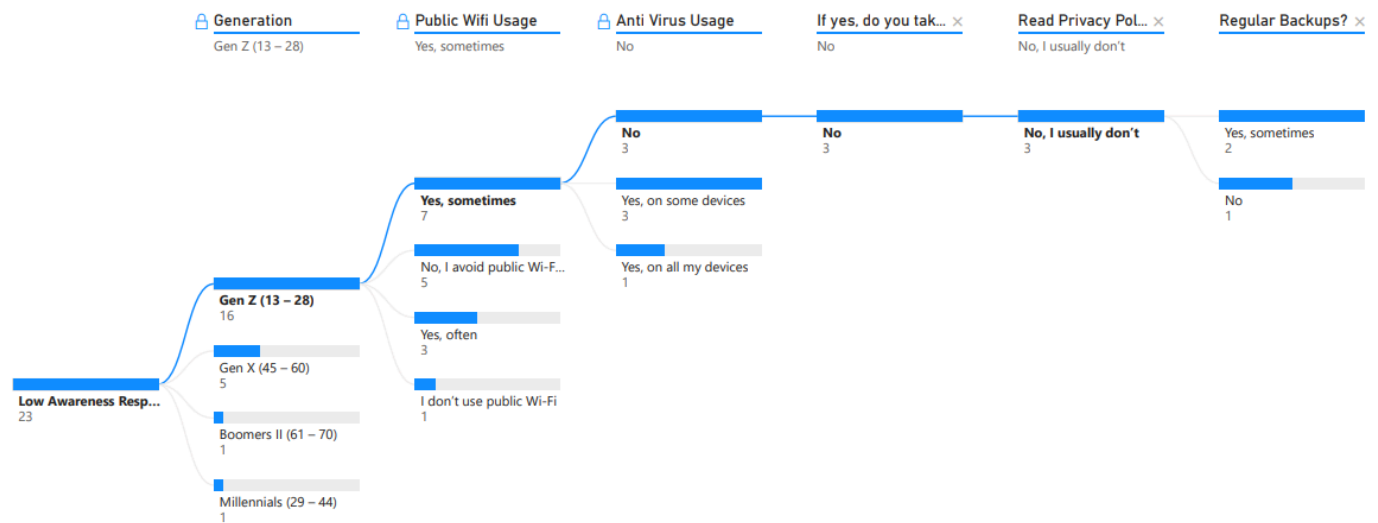


Figure 3

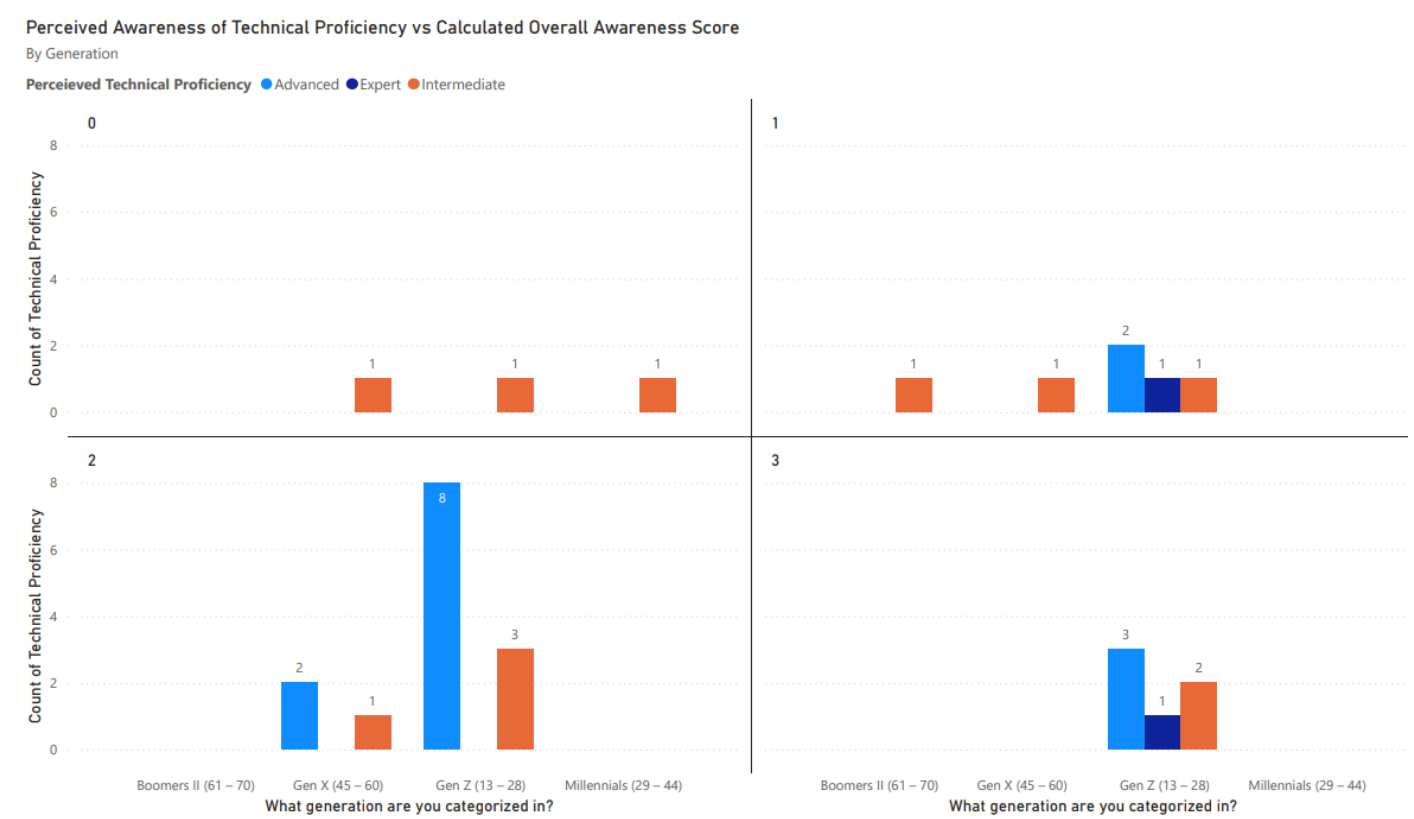


Figure 4

Data Protection Rights Awareness by Generation				
Generation	No	Unsure	Yes	Total
Gen Z (13 – 28)	15	5	2	22
Gen X (45 – 60)	3	2		5
Boomers II (61 – 70)		1		1
Millennials (29 – 44)	1			1
Total	19	8	2	29

Figure 5

Open Coding

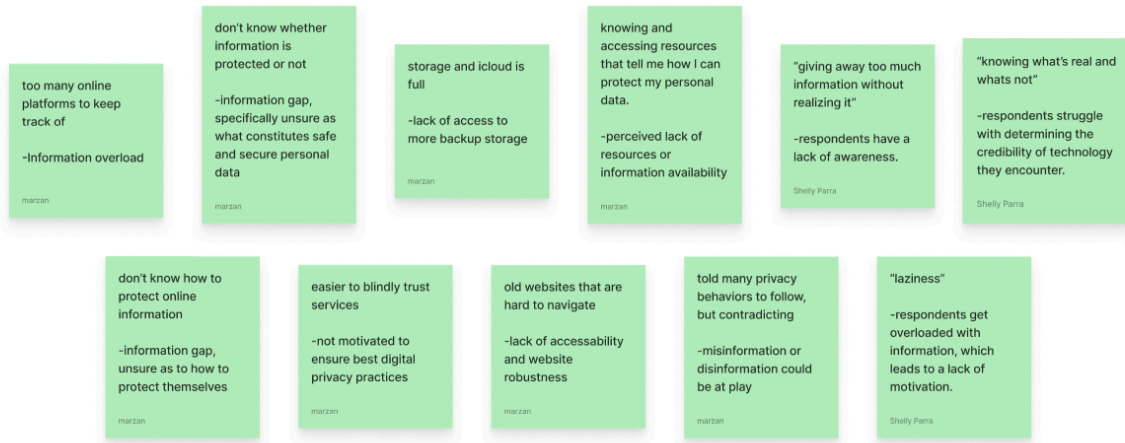


Figure 6

Clustering (Axial Coding)

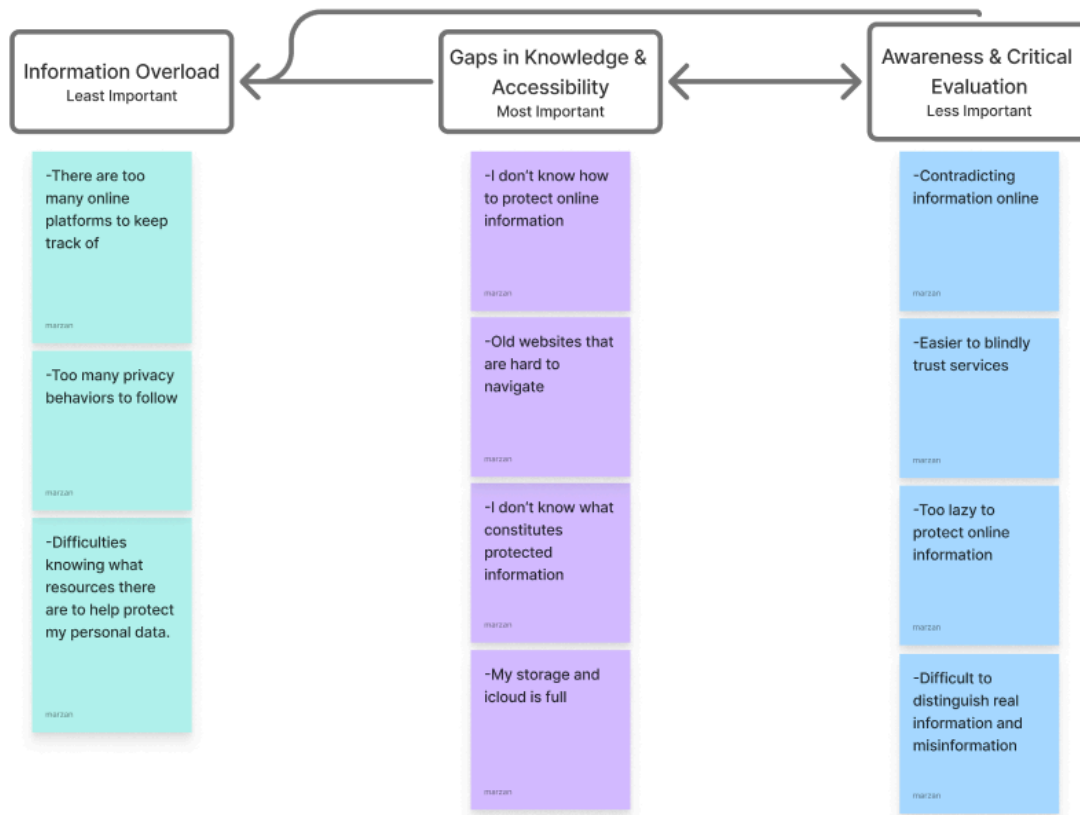
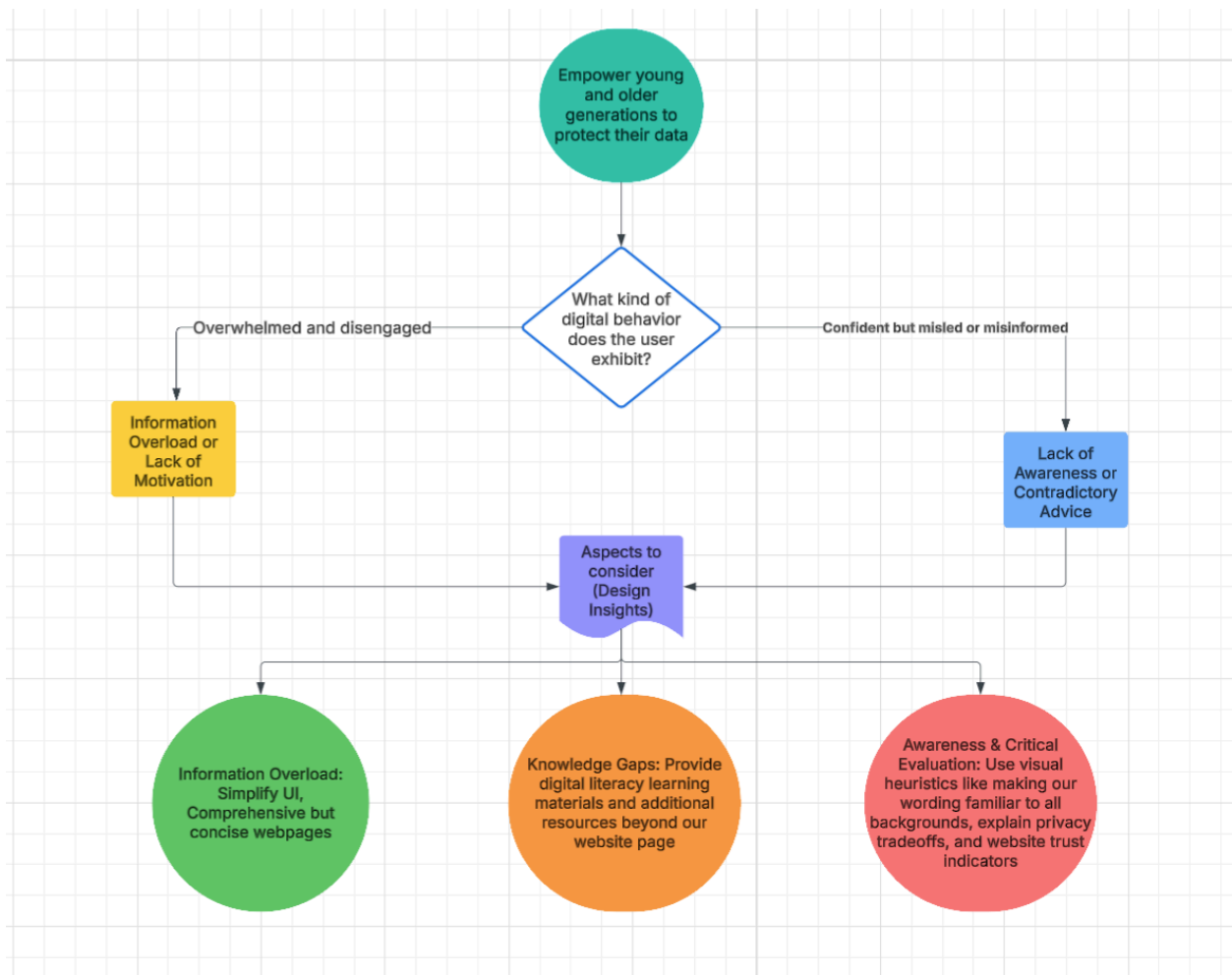


Figure 7

Affinity Diagram



Conclusion

By combining behavioral survey data with coded qualitative feedback, we were able to uncover what users are doing online as well as why they engage in certain behaviors. Our quantitative analysis—using tools like a decomposition tree and a calculated cybersecurity awareness score—highlighted key patterns of risk, such as users who regularly use public Wi-Fi for sensitive tasks without employing protective tools like VPNs or antivirus software. These findings were put into context through our qualitative coding methods and affinity diagram. The qualitative analysis revealed underlying motivations and barriers, such as confusion, being overwhelmed, or misplaced confidence.

A consistent theme that emerged across our data is the disconnect between users' perceived proficiency and their actual cybersecurity awareness. To illustrate, many Gen Z respondents rated themselves as advanced yet scored low on awareness indicators like knowing their data protection rights or recognizing digital threats. This signals that being digitally

literate does not necessarily equate to being digitally secure. It also reinforces the importance of educational interventions, like DataGear, that present the best practices.

Through this integration, we developed personas such as the overwhelmed beginner, who lacks both motivation and clarity, and the curious but misinformed user, who feels confident but unknowingly engages in risky behaviors. As we reflect on our findings, one area for further exploration is how to best translate abstract cybersecurity concepts into content that feels relevant and digestible for non-expert users. Overall, our early research has identified key knowledge gaps; the challenge ahead is determining how to frame this information properly.